
DSC 291 Trustworthy ML

Course Proposal Template (Research Reproduction)

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Project format: Format 1 — Tutorial on Research Reproduction.

Paper: *JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models* (NeurIPS 2024) (1).

1 Introduction

We plan to use **Format 1: Tutorial on Research Reproduction** to reproduce a scoped version of *JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models*. Large language models are now deployed as the backbone of user-facing assistants and tool-using agents, so their refusal behavior must hold under adversarial inputs, not just benign ones. A jailbreak attack attempts to bypass a model’s safety alignment and elicit harmful or policy-violating responses. Evaluating jailbreak robustness is hard because papers vary in model versions, prompt sets, attack artifacts, and judges — making cross-paper comparison and reproduction unreliable.

JailbreakBench responds with four concrete artifacts: a fixed dataset of 100 misuse and 100 benign behaviors aligned to provider usage policies, a versioned repository of jailbreak strings (PAIR, GCG, and others), a standardized judge specification, and a public leaderboard. This paper is closely related to DSC 291 because it studies robustness in NLP and AI safety: whether aligned models can maintain safe refusal behavior under adversarial prompting, and how such robustness should be measured reproducibly. Because the benchmark provides public artifacts and clear metrics such as attack success rate and refusal/refusal-like behavior, it is suitable for a controlled reproduction project.

2 Reproduction Plan

2.1 Results, Tables, and Figures We Plan to Reproduce

We will reproduce a **scoped subset** of the JailbreakBench evaluation pipeline, targeting the paper’s core attack-method ranking rather than full leaderboard numbers. Specifically, we plan to reproduce:

1. **Attack Success Rate (ASR) comparison** across attack methods (e.g., PAIR and GCG artifacts released with the benchmark) on our selected target models, mirroring Table 1 of Chao et al. (2024), which reports ASR by attack method and target model.
2. **Refusal rate on benign behaviors**, evaluating over-refusal and the utility–safety trade-off, corresponding to the paper’s benign-refusal-rate analysis (Section 4 of Chao et al., 2024).
3. **Defense comparison**, evaluating at least one defense method (e.g., SmoothLLM or perplexity-based filtering) against the same attack artifacts.
4. **Qualitative examples** showing representative successful jailbreaks, failed jailbreaks, and ambiguous judging cases that expose evaluator limitations.

2.2 Scope and Implementation Plan

Target models. 2–3 models, sampling across openness and provider: one open-weight model (candidate: Llama-3-8B-Instruct or Vicuna-7B) for local/Colab inference, and one closed-source API model (candidate: GPT-4o-mini or Claude Haiku) chosen for cost control.

Behavior subset. A stratified sample of 30–50 prompts from JBB-Behaviors covering all harm categories, plus the matched benign counterparts.

Judge. We will use the official Llama-Guard-2 judge as our primary evaluator. We will additionally spot-check with a simple refusal-string heuristic to characterize judge sensitivity.

Workflow. (1) Install and verify the jailbreakbench package and access the artifact repository; (2) sample the behavior subset; (3) run the benchmark pipeline to collect model responses on selected attack artifacts; (4) score with Llama-Guard-2; (5) compute ASR and refusal-rate tables; (6) compile qualitative examples; and (7) write the Medium tutorial documenting setup, results, and reproduction difficulties.

2.3 Additional Experiments

Beyond the core reproduction, we plan two extensions:

1. **Judge sensitivity.** We will re-score a subset of responses using multiple judges (Llama-Guard-2, GPT-4 as judge, and a rule-based refusal-string heuristic) to quantify how the measured ASR shifts with judge choice—directly probing a known limitation of jailbreak benchmarks.
2. **Model version sensitivity.** For the closed-source API model, we will run the same prompt subset at two different time points to measure how reproducible the reported ASR is across silent provider updates, which the original paper does not study.

2.4 Risks and Mitigations

We anticipate three main risks. (a) API cost can grow with judge queries; we will cap this by sampling and by using Llama-Guard-2 (free) as the primary judge. (b) Jailbreak content is safety-sensitive; we will not release new jailbreak artifacts, will use only artifacts already public in the benchmark, and will frame the blog around evaluation methodology rather than attack craft. (c) Repository dependencies may not install cleanly on Colab; we will validate the end-to-end install in the first week and document any patches.

References

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- [2] P. Chao, A. Robey, E. Dobriban, H. Hassani, G. J. Pappas, E. Wong. *Jailbreaking Black Box Large Language Models in Twenty Queries*. arXiv:2310.08419, 2023.
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- [5] H. Inan et al. *Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations*. arXiv:2312.06674, 2023.